University of Zurich[UZH]

Graph Neural Networks for Epidemic Source Detection on Temporal Contact Networks

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Zusammenfassung

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Abstract

Identifying the origin of an epidemic outbreak—the so-called patient-zero or source detection problem—is a fundamental challenge in network epidemiology. Given the observed infection state of a contact network at some time T, the task is to rank network nodes by their likelihood of having initiated the spreading process. While graph neural network (GNN) methods have recently demonstrated competitive performance on this task, the majority of existing approaches operate on static projections of the contact graph, thereby discarding the temporal ordering of contacts that governs causal transmission paths.

This thesis addresses that gap by implementing and systematically comparing four GNN architectures of increasing temporal expressiveness for source detection on temporal contact networks. The evaluated models are: (i) a static GNN baseline that processes the time-aggregated graph; (ii) the Backtracking Network (BN) of [**?**], which encodes temporal activation patterns as edge textures; (iii) the De Bruijn Graph Neural Network (DBGNN) of [**?**], which constructs higher-order De Bruijn graphs to represent causal walks; and (iv) a DAG-GNN applied to Temporal Event Graphs, combining the event-graph representation with directed acyclic graph convolutions [**?**]. All models are trained in a supervised fashion using ground-truth sources generated by Monte Carlo SIR simulations on synthetic and real-world contact networks via an adapted version of Holme's fast temporal SIR engine.

The thesis provides what is, to our knowledge, the first systematic multi-architecture comparison of temporal GNN methods for epidemic source detection. We examine three research hypotheses concerning causal path integrity, temporal granularity convergence, and signal-to-noise resilience in late-stage outbreaks, thereby contributing both empirical benchmarks and architectural insights to the emerging field of learning-based network epidemiology.

Contents

1 Introduction **1**
1.1 Motivation . . . 1
1.2 Background and the Gap in Existing Work . . . 1
1.3 This Thesis . . . 2
1.4 Contributions . . . 3
1.5 Thesis Structure . . . 3

2 Foundations **5**
2.1 Network Science and Graph Theory . . . 5
2.1.1 Static vs. Temporal Networks . . . 5
2.1.2 Causality and Causal Walks in Dynamic Graphs . . . 6
2.2 Epidemic Spreading Models . . . 6
2.2.1 The SI, SIR, SIS, and SIRS Models . . . 7
2.2.2 The SIR Model and Stochastic Transitions . . . 7
2.3 Classical Source Detection Algorithms . . . 8
2.3.1 Rumor Centrality and Jordan Center . . . 8
2.3.2 Simulation-based Methods: Soft Margin Estimator (SME) . . . 8
2.4 Graph Neural Networks (GNNs) . . . 9
2.4.1 Message Passing Paradigm . . . 9
2.4.2 GNNs for Static Source Detection (GCN, GraphSAGE, GAT) . . . 10
2.4.3 GNNs for Temporal Source Detection . . . 10

3 Related Work **11**
3.1 Classical Source Detection Methods . . . 11
3.2 GNN-Based Methods on Static Networks . . . 12
3.3 GNN-Based Methods on Temporal Networks . . . 13
3.4 Summary and Research Gap . . . 14

4 Methodology: GNNs for Temporal Source Detection **15**
4.1 Formalization of the Backtracking Problem . . . 15
4.2 Data Generation via Monte Carlo Simulations . . . 16
4.3 Representation of Temporal Information . . . 17
4.3.1 Static Projection (Baseline) . . . 17

4.3.2 Temporal Discretization (Timeslices) 18
4.3.3 Edge Feature Encoding for Timestamps 18

5 Model Architectures and Implementation 21
5.1 Evaluated GNN Models 21
5.1.1 Static Projection (Baseline) 21
5.1.2 De Bruijn Graph Neural Networks (DBGNN) 22
5.1.3 Kernel-based GNN / Backtracking Network (BN) 23
5.1.4 Directed Acyclic Graph GNN (DAG-GNN) on Temporal Event Graphs 26
5.2 Hyperparameter Optimisation and Training Setup 28

6 Experimental Evaluation 29
6.1 Datasets 29
6.1.1 Synthetic Graphs 29
6.1.2 Real-world Contact Networks 30
6.2 Evaluation Metrics 30
6.2.1 Top-k Accuracy 30
6.2.2 Average Error Distance and Reciprocal Rank 31
6.2.3 Resistance Score for Uncertainty Quantification 31
6.3 Results and Analysis 32
6.3.1 Performance Comparison: GNNs vs. Classical Heuristics 32
6.3.2 Impact of Network Density and Observation Time T 32
6.3.3 Robustness to Missing Information (Links/Node States) 33

7 Discussion 35
7.1 Interpretation of Results 35
7.2 Advantages of Temporal Architectures over Static Models 36
7.3 Ablation Studies 37

8 Limitations 39

9 Future Work 41

10 Conclusions 43
10.1 Summary of Contributions 43
10.2 Limitations 44
10.3 Future Work 44

A Appendix 47
A.1 Lorem Ipsum 47

1

Introduction

1.1 Motivation

Identifying the origin of an epidemic—the so-called "patient zero"—is one of the central problems at the intersection of network science and public health. When a pathogen emerges, reconstructing its point of introduction can inform containment strategies, guide contact-tracing efforts, and ultimately limit the scale of an outbreak. The societal stakes are high: the COVID-19 pandemic demonstrated that delays in pinpointing an epidemic's geographic and individual origin translate directly into uncontrolled community transmission [**?**]. Similar imperatives arise in veterinary epidemiology, where the rapid identification of index farms during foot-and-mouth disease outbreaks has been shown to be critical for preventing national-scale livestock crises [**?**]. Beyond infectious disease, structurally analogous source-detection problems appear in the spread of misinformation and rumours on online social networks, where tracing a viral piece of content back to its origin is both a technical and a social challenge [**?**].

The increasing availability of high-resolution contact data has made computational approaches to this problem tractable. Bluetooth proximity sensors in hospital settings, RFID badges in conference environments, and digitised livestock movement records all yield datasets in which individual contacts are timestamped to the minute or second. These temporal contact networks carry strictly more information than the static projections traditionally used in network epidemiology: the ordering and timing of contacts governs which transmission chains are causally possible, a fact that static aggregation systematically discards.

1.2 Background and the Gap in Existing Work

Epidemic processes are naturally modelled as spreading processes on networks. An infected node transmits the pathogen to its susceptible neighbours according to a stochastic process; the outcome is an observed snapshot of infected, recovered, and susceptible individuals. Given only this final snapshot and the underlying contact structure, the task is to infer the node most likely to have initiated the outbreak. Early analytical work by Shah and Zaman introduced rumour centrality as a maximum likelihood estimator

for the source of a spreading process on tree-structured networks [**?**]. This sparked a productive tradition: the Jordan centre was proposed as a tractable approximation for general network topologies [**?**], dynamic message-passing (DMP) extended the framework to arbitrary spreading parameters [**?**], and the Soft Margin Estimator (SME) introduced a simulation-based approach that quantifies source likelihood by comparing observed infection snapshots with Monte Carlo realisations [**?**]. Loopy belief propagation offered a probabilistically principled alternative for networks with cycles [**?**].

Despite the empirical availability of temporal contact data, the overwhelming majority of these methods were designed for static networks, treating the contact structure as a fixed graph. Temporal networks—which record the precise timing and ordering of contacts—are a more realistic substrate. Several methods have been adapted or developed for this setting: Lokhov et al. extended dynamic message-passing to sequences of contact snapshots [**?**], and Rocha and Masuda proposed individual-based approximations for temporal spreading processes [**?**]. Nevertheless, the structured temporal ordering of contacts—which determines the set of causally valid transmission paths—has largely been treated as a second-order consideration rather than a primary modelling target.

Graph Neural Networks (GNNs) have recently emerged as a powerful and data-driven approach to source detection, reframing the problem as supervised classification trained on simulated epidemic outcomes [**?**, **?**]. A comprehensive review and benchmark by Sterchi, Brack, and Hilfiker established that GNN-based methods substantially outperform classical analytical and simulation-based baselines across a range of network topologies and epidemic parameters [**?**]. Critically, however, that study also identifies a fundamental limitation of the field: virtually all GNN methods for source detection operate on static projections of the contact network, collapsing temporal information before any learning takes place. Sterchi et al. explicitly note that Ru et al.'s Backtracking Network is the sole published exception [**?**]. There is currently no systematic study comparing multiple temporal GNN architectures, each embodying a distinct inductive bias about how temporal structure should be represented, against each other and against both static GNN and classical baselines on temporal contact networks.

1.3 This Thesis

This thesis addresses that gap by developing and evaluating three GNN architectures specifically designed to exploit temporal contact information for epidemic source detection. The first is the **Backtracking Network (BN)** proposed by Ru et al. [**?**], which operates on the aggregated static graph augmented with binary edge activation patterns—binary vectors recording at which time steps each contact was active—thereby encoding temporal structure as edge features while preserving the scalability of standard graph convolution. The second is the **De Bruijn Graph Neural Network (DBGNN)** proposed by Qarkaxhija et al. [**?**], which constructs higher-order De Bruijn graph representations to explicitly model causal walks and non-Markovian temporal dynamics that cannot be captured by first-order aggregation. The third is a **DAG-GNN on Temporal Event Graphs (TEG)**, combining the event-graph representation of Saramäki

et al. [**?**]—which maps a temporal network into a directed acyclic graph whose nodes are contact events and whose edges encode time-respecting paths—with the DAG convolutional network (DCN) of Rey et al. [**?**], designed to perform spectral convolution on such acyclic structures. A static GNN operating on the time-aggregated contact graph serves as the primary baseline.

Training data for all models is generated via Monte Carlo SIR simulations using an adapted version of Holme's fast C-based temporal SIR simulation engine [**?**], and all experiments are tracked using the Weights and Biases framework.

Three research hypotheses guide the experimental evaluation. **H1 (Causal Path Integrity)** posits that temporal GNN architectures provide the highest marginal accuracy gain in structured networks where temporal ordering strictly limits the reachable set of nodes. **H2 (Granularity Convergence)** posits that the advantage of temporal architectures diminishes as the temporal resolution of the input decreases, with accuracy converging toward the static baseline as time-bin width increases. **H3 (Temporal Signal-to-Noise Resilience)** posits that temporal GNNs maintain higher accuracy than static models in late-stage outbreaks by pruning the combinatorial explosion of plausible transmission paths present in the aggregated contact history.

1.4 Contributions

This thesis makes the following contributions. First, we provide a unified re-implementation of the Backtracking Network of Ru et al. within a common experimental framework, enabling controlled comparison. Second, we adapt the De Bruijn GNN of Qarkaxhija et al. and the DAG convolutional network of Rey et al. to the source detection task, which has not been done in prior work. Third, we conduct the first systematic multi-architecture comparison of temporal GNN methods for source detection against each other and against classical baselines under identical data and evaluation conditions. Fourth, we analyse the robustness of each architecture to missing link observations and partial node-state information.

1.5 Thesis Structure

Chapter 2 provides the necessary background in network science, epidemic spreading models, and graph neural networks. Chapter 3 surveys related work, covering classical analytical methods, simulation-based approaches, static GNN methods, and the limited existing literature on temporal GNN source detection. Chapter 4 formalises the backtracking problem and describes the data generation pipeline. Chapter 5 presents the three model architectures and their implementation. Chapter 6 describes the experimental setup and datasets. Chapter 7 reports and analyses results. Chapter 8 discusses interpretations, limitations, and directions for future work. Chapter 9 concludes.

2

Foundations

This chapter introduces the formal and conceptual background required for the remainder of the thesis. We begin with the mathematical foundations of network science and distinguish static from temporal contact networks, emphasising the causal structure that temporal ordering imposes on spreading processes. We then review the family of compartmental epidemic models, with particular attention to the stochastic SIR model that underpins all experiments in this work. Classical source detection algorithms are surveyed next, establishing the non-machine-learning baselines against which the proposed methods are later evaluated. The chapter concludes with a self-contained introduction to graph neural networks, covering the general message-passing paradigm, its principal static instantiations, and the challenges that arise when extending these methods to temporal network data.

2.1 Network Science and Graph Theory

A graph $G = (\mathcal{V}, \mathcal{E})$ consists of a finite set of nodes $\mathcal{V}$ with $|\mathcal{V}| = N$ and a set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. In an undirected graph, each edge $\{u, v\} \in \mathcal{E}$ represents a symmetric relationship; in a directed graph, edges are ordered pairs (u, v) indicating a relationship from u to v. The topology of a graph is compactly encoded by its adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, where $A_{uv} = 1$ if and only if $(u, v) \in \mathcal{E}$. For undirected graphs, $\mathbf{A}$ is symmetric. The degree of a node v, denoted d_v, counts the number of edges incident to it; for directed graphs, one distinguishes the in-degree d_v^{in} and out-degree d_v^{out}. The neighbourhood of v is the set $\mathcal{N}(v) = \{u \in \mathcal{V} : (u, v) \in \mathcal{E}\}$ of nodes adjacent to v. Nodes may carry feature vectors $\mathbf{x}_v \in \mathbb{R}^d$, and edges may carry weights or attributes; these attributed graphs form the primary input to the GNN models studied in this thesis.

2.1.1 Static vs. Temporal Networks

A static network treats the edge set $\mathcal{E}$ as fixed over time. This is an appropriate abstraction for many structural or relational datasets, but it is too coarse for systems in which interactions are transient and sequentially ordered. A temporal network, by contrast, is a triple $\mathcal{G} = (\mathcal{V}, \mathcal{E}^t, \mathcal{T})$, where $\mathcal{T} = \{t_1, t_2, \ldots, t_M\}$ is an ordered set of event times and each edge $(u, v, t) \in \mathcal{E}^t$ records a contact between nodes u and v occurring at time t. Temporal networks may equivalently be represented as a sequence of graph snapshots

$\{G_0, G_1, \ldots, G_T\}$, where $G_t = (\mathcal{V}, E_t)$ and E_t contains all contacts active during the time interval $[t, t+1)$.

The aggregated static graph $G_a = (\mathcal{V}, E_a)$ is obtained by taking the union of all snapshot edge sets, $E_a = \bigcup_{t=0}^{T} E_t$. While G_a records which pairs of nodes were ever in contact, it discards the temporal ordering entirely. This loss of information is consequential for epidemic spreading: a contact $(A, B, t = 1)$ followed by $(B, C, t = 2)$ permits the transmission chain $A \to B \to C$, whereas the reversed sequence $(B, C, t = 1)$ followed by $(A, B, t = 2)$ does not, because B cannot have been infected by A before the contact between them occurred. The static projection G_a treats both orderings identically. Real-world temporal contact networks arise in many domains, including face-to-face proximity measured by RFID or Bluetooth sensors, digitised livestock movement records, and timestamped email or messaging exchanges.

2.1.2 Causality and Causal Walks in Dynamic Graphs

The causal structure of a temporal network is captured by the notion of a time-respecting path, also called a causal walk. Formally, a causal walk from node v_0 to node v_k is a sequence of contacts $((v_0, v_1, t_1), (v_1, v_2, t_2), \ldots, (v_{k-1}, v_k, t_k))$ such that all contacts belong to $\mathcal{E}^t$ and the event times are strictly increasing, $t_1 < t_2 < \cdots < t_k$. The existence of a causal walk from v_0 to v_k is the necessary condition for a pathogen originating at v_0 to reach v_k through sequential transmission.

The reachable set $\mathcal{R}(v, T)$ from a source node v by time T is defined as the set of all nodes w for which a causal walk from v to w exists with $t_k \leq T$. This set can be strictly smaller than the reachable set on the static projection G_a, particularly in networks with bursty or structured contact patterns. For source detection, the causal path structure is the fundamental quantity of interest: the set of candidate sources for an observed infected node i at time T is precisely the set of nodes from which a causal walk leading to i exists before T. Methods that operate on G_a rather than $\mathcal{G}$ implicitly treat both causally valid and invalid transmission chains as equally plausible, introducing false candidates into the inference problem. Exploiting causal walk structure is therefore not merely a modelling refinement but a logical necessity for exact source inference on temporal networks.

2.2 Epidemic Spreading Models

Compartmental models are the dominant framework for mathematical epidemiology. They partition the population into a small number of discrete states and specify probabilistic rules governing transitions between states. The simplest models are fully determined by two parameters: a transmission probability β governing the rate at which the pathogen spreads from infected to susceptible individuals, and a recovery probability μ governing the rate at which infected individuals recover. We describe the principal compartmental models in order of increasing complexity before specialising to the SIR model used throughout this thesis.

2.2.1 The SI, SIR, SIS, and SIRS Models

The SI model is the simplest epidemic model, containing only two compartments: susceptible (S) and infectious (I). An infected node transmits to each susceptible neighbour with probability β per contact; there is no recovery. The entire connected component of the source eventually becomes infected, making SI appropriate only for processes where immunity or intervention is absent.

The SIS model extends SI by allowing recovery: an infected node transitions back to the susceptible state with probability μ per time step, reflecting the assumption that infection confers no lasting immunity. This produces an endemic equilibrium in which the pathogen persists within the population, and it is the standard model for recurrent illnesses such as the common cold. The SIR model introduces a third compartment, recovered (R): an infected node transitions to R with probability μ, and recovered nodes are permanently immune and cannot be reinfected. The SIR model is appropriate for acute infectious diseases such as influenza or measles and is the primary model used in this thesis. The SIRS model allows immunity to wane over time, so that recovered nodes return to the susceptible pool with probability ω per time step, enabling recurrent epidemic waves. For the purposes of this work, the irreversibility of the recovered state in the SIR model is a key property: the final observed snapshot of SIR states constitutes a permanent record of which nodes were ever infected, simplifying the inference problem.

2.2.2 The SIR Model and Stochastic Transitions

We define the stochastic SIR process on a temporal contact network $\mathcal{G}$. Each node $v \in \mathcal{V}$ occupies a state $X_v(t) \in \{S, I, R\}$ at each discrete time step t. The process is initialised by setting $X_s(0) = I$ for the source node s and $X_v(0) = S$ for all $v \neq s$. Two types of stochastic transition are applied at each time step. First, for each contact $(i, j, t) \in \mathcal{E}^t$ where $X_i(t) = I$ and $X_j(t) = S$, the susceptible node j becomes infected with probability β:

$$\mathbb{P}\big(X_j(t+1) = I \mid X_j(t) = S,\ X_i(t) = I,\ (i, j, t) \in \mathcal{E}^t\big) = \beta.$$

Second, each infected node i with $X_i(t) = I$ independently recovers with probability μ:

$$\mathbb{P}(X_i(t+1) = R \mid X_i(t) = I) = \mu.$$

Recovered nodes remain in state R for all subsequent time steps. The process terminates when no infected nodes remain.

In this thesis, β and μ are fixed hyperparameters specified in experiment configuration files. Different values of (β, μ) produce outbreaks of different expected size and duration: high β relative to μ yields large outbreaks where many nodes are infected, while low β or high μ produces small, rapidly extinguished outbreaks. Training data is generated by running the SIR process repeatedly from each candidate source, with independent random draws for all transition events at each repetition. This Monte Carlo procedure, implemented using an adapted version of Holme's fast C-based temporal SIR simulation engine [**?**], produces large ensembles of simulated epidemic outcomes that form the supervised training and evaluation datasets for all models in this thesis.

2.3 Classical Source Detection Algorithms

Before the advent of machine learning approaches to source detection, the field relied on two complementary families of methods: analytical estimators derived from probabilistic first principles, and simulation-based estimators that score candidate sources by comparing observed and simulated outbreak snapshots. We summarise the most relevant representatives of each family, as they constitute the classical baselines evaluated in Chapter 6.

2.3.1 Rumor Centrality and Jordan Center

Shah and Zaman [**?**] established the theoretical foundation of algorithmic source detection by showing that the source of a spreading process on a regular tree can be identified as a maximum likelihood estimator under the SI model. They defined the rumour centrality $R(v)$ of a node v as the number of orderings in which the observed infection pattern can arise if v is the source, and proved that the node maximising $R(v)$—the rumour centre—is a strongly consistent estimator of the true source as the network size grows. The Jordan centre extends this idea to general network topologies and the SIR model: it is defined as the node that minimises the maximum shortest-path distance to any infected or recovered node, $v^* = \arg\min_{v \in \mathcal{V}_{I \cup R}} \max_{u \in \mathcal{V}_{I \cup R}} d(v, u)$, where $d(\cdot, \cdot)$ denotes the shortest path on the induced infected subgraph. Both estimators are computationally inexpensive and require no knowledge of the spreading parameters β or μ. Their principal limitation is a reliance on tree-structure approximations and shortest-path distances on the static projection, which means they cannot exploit the temporal ordering of contacts or account for cycles in the contact network.

2.3.2 Simulation-based Methods: Soft Margin Estimator (SME)

The Soft Margin Estimator (SME) of Antulov-Fantulin et al. [**?**] takes a Monte Carlo approach to source scoring. For each candidate source v, M independent SIR simulations are run from v on the contact network, producing a collection of simulated final snapshots $\{\tilde{r}_{v,m}\}_{m=1}^{M}$. The SME score for source v is then the mean similarity between the observed snapshot r^* and each simulated snapshot, measured by a soft-margin kernel:

$$P_{\text{SME}}(v) \propto \frac{1}{M} \sum_{m=1}^{M} \exp\Big(- \left[\varphi(r^*, \tilde{r}_{v,m}) - 1\right]^2 \Big) ,$$

where $\varphi(r^*, \tilde{r}_{v,m})$ is the Jaccard similarity between the observed and simulated infected-plus-recovered sets. The SME is model-agnostic in the sense that any stochastic spreading process can be substituted in place of SIR, and it naturally handles partial observations by comparing only those nodes for which state information is available. Its principal weakness is computational cost: scoring all N candidate sources requires $N \times M$ full epidemic simulations, which is prohibitive for large networks and typically demands $M \geq 1000$ simulations per candidate to achieve stable score estimates. As noted by

Sterchi et al. [**?**], SME is frequently used as a gold-standard classical baseline in benchmarks of GNN-based methods precisely because it is the closest classical analogue to the simulation-based training signal used by supervised neural approaches.

2.4 Graph Neural Networks (GNNs)

Graph neural networks are a family of architectures that generalise deep learning to graph-structured data by defining differentiable operations that respect the topology of the input graph. Unlike convolutional neural networks, which exploit the regular grid structure of image data, GNNs must be permutation-equivariant with respect to node ordering and must aggregate information across irregular, variable-degree neighbourhoods. The message-passing framework provides the canonical abstraction for this class of models.

2.4.1 Message Passing Paradigm

The general message-passing neural network (MPNN) framework, formalised by Gilmer et al. [**?**], describes GNN computation in terms of two alternating operations applied at each layer $l = 1, \ldots, L$. In the aggregation step, each node v collects messages from its neighbours:

$$\mathbf{m}_v^{(l)} = \mathrm{AGG}\Big(\Big\{ M_l\Big(\mathbf{h}_v^{(l-1)},\, \mathbf{h}_u^{(l-1)},\, \mathbf{e}_{uv}\Big) : u \in \mathcal{N}(v) \Big\}\Big)\,,$$

where M_l is a learnable message function, $\mathbf{e}_{uv}$ is an optional edge feature, and AGG is a permutation-invariant aggregator such as sum, mean, or max. In the update step, the node's own embedding is combined with the aggregated message:

$$\mathbf{h}_v^{(l)} = U_l\Big(\mathbf{h}_v^{(l-1)},\, \mathbf{m}_v^{(l)}\Big)\,.$$

Node embeddings are initialised from input features, $\mathbf{h}_v^{(0)} = \mathbf{x}_v$. After L layers of message passing, the embedding $\mathbf{h}_v^{(L)}$ encodes a summary of the L-hop neighbourhood of v. Different architectural choices for M_l, AGG, and U_l yield the principal GNN variants. The Graph Convolutional Network (GCN) of Kipf and Welling [**?**] uses a symmetric degree-normalised aggregation without a distinct message function:

$$\mathbf{h}_v^{(l)} = \mathrm{ReLU}\left(\mathbf{W}^{(l)} \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{\mathbf{h}_u^{(l-1)}}{\sqrt{\tilde{d}_v\, \tilde{d}_u}}\right),$$

where $\tilde{d}_v = d_v + 1$ accounts for the added self-loop. GraphSAGE [**?**] concatenates the node's own embedding with the mean of its neighbours' embeddings before applying a linear transformation, making it well-suited for inductive settings. The Graph Attention Network (GAT) of Veličković et al. [**?**] replaces fixed normalisation with learned attention coefficients, allowing the model to weight neighbour contributions dynamically based on feature similarity. All three variants are evaluated as static baselines in this thesis.

2.4.2 GNNs for Static Source Detection (GCN, GraphSAGE, GAT)

The source detection problem on a static network can be formulated as a node classification task: given an attributed graph $(G_a, \mathbf{X})$ where $\mathbf{X} \in \{0,1\}^{N\times 3}$ encodes the one-hot SIR state of each node at the time of observation, predict which node is the epidemic source. GNNs are particularly well-suited to this task because source nodes tend to lie near the topological centre of the infected subgraph, and iterative neighbourhood aggregation naturally propagates the local density of infected nodes inward toward the source. This intuition is supported empirically by the benchmark of Sterchi et al. [**?**], who show that GCN, GraphSAGE, and GAT all substantially outperform classical methods including the Jordan centre and the SME across a range of network topologies. Training is performed in a supervised manner on a large collection of simulated epidemic outcomes: each training sample is a triple consisting of the contact graph, the node-state feature matrix, and the identity of the true source node. The loss is the cross-entropy between the softmax output distribution over nodes and the one-hot ground-truth source label. The softmax formulation is important because it frames source detection as a competition among nodes, whereas a sigmoid output would treat each node's probability independently and is inappropriate for the single-source setting.

2.4.3 GNNs for Temporal Source Detection

Static GNNs applied to the aggregated graph G_a are inherently limited by the loss of temporal ordering: two nodes may have identical degree and neighbourhood structure in G_a while having very different causal reachability sets in the temporal network $\mathcal{G}$. The goal of temporal GNN architectures is to encode the time-varying contact structure in a way that is informative for source inference. Three broad strategies have emerged in the literature and are evaluated in this thesis. The first strategy, employed by the Backtracking Network of Ru et al. [**?**], augments the edges of the aggregated graph with binary activation vectors—referred to as edge textures—that record at which discrete time steps each contact was active; this preserves the scalability of standard graph convolution while providing the model with access to temporal co-occurrence patterns (Section 5.1.3). The second strategy, used by the De Bruijn Graph Neural Network of Qarkaxhija et al. [**?**], lifts the temporal network into a higher-order state space whose nodes represent ordered sequences of contacts, thereby encoding causal walk structure as an explicit graph topology on which standard message passing can operate (Section 5.1.2). The third strategy combines the Temporal Event Graph representation of Saramäki et al. [**?**]—which maps the temporal network into a directed acyclic graph whose nodes are contact events and whose edges encode time-respecting sequential relationships—with the DAG convolutional network of Rey et al. [**?**], enabling spectral learning directly on the acyclic causal structure (Section 5.1.4). Each strategy encodes a different inductive bias about how temporal information should be represented and aggregated, and a primary contribution of this thesis is a systematic comparison of the detection accuracy achieved by each approach under identical experimental conditions.

3

Related Work

The source detection problem has attracted substantial attention across several research communities over the past fifteen years. Work in this area can be grouped into three broad traditions: classical analytical and simulation-based methods that derive estimators from probabilistic first principles; supervised machine learning methods—primarily GNN-based—that operate on static projections of the contact network; and a nascent body of work that attempts to exploit temporal structure more directly. We survey each tradition in turn and conclude by identifying the research gap that this thesis addresses.

3.1 Classical Source Detection Methods

The foundational contribution to algorithmic source detection is the work of Shah and Zaman [**?**], who observed that the source of a spreading rumour on a tree-structured network can be identified as a maximum likelihood estimator under the SI model. They defined rumour centrality as the number of ways the observed infection pattern can be explained as a spreading process rooted at a given node, and showed that the node maximising this quantity—the rumour centre—is a strongly consistent estimator of the true source as network size grows. An extension to directed and more general network topologies, the Jordan centre, selects the node that minimises the maximum distance to any infected node, approximating the centroid of the infection subgraph. Both estimators are computationally efficient and do not require knowledge of the spreading parameters, but they inherit a fundamental limitation: the tree-structure assumption fails on networks with cycles, and neither method accounts for the temporal ordering of contacts.

Lokhov et al. proposed dynamic message-passing (DMP) as a principled approach to approximate the marginal probabilities of infection under the SIR model on arbitrary graphs [**?**]. DMP iterates a system of equations that propagates probability messages along edges, and the inferred source is the node whose initialisation as infected maximises the likelihood of the observed snapshot. The method is fast and scales to moderate network sizes, but it requires precise knowledge of the transmission probability β and recovery probability μ, which are rarely known in practice, and it becomes inaccurate on networks with many short cycles due to the Bethe approximation underlying the message-passing equations.

The Soft Margin Estimator (SME) of Antulov-Fantulin et al. [**?**] takes a different approach: for each candidate source, a large ensemble of Monte Carlo SIR simulations is run, and the source score is defined as the proportion of simulations whose final snapshot is compatible with the observed one under a soft-margin criterion. The method is model-agnostic in the sense that any stochastic spreading model can be plugged in, and it is the only classical method that naturally handles partial observations. Its principal drawback is computational cost: a reliable score requires thousands of simulations per candidate node, yielding quadratic scaling in the number of candidate sources. Braunstein and Ingrosso [**?**] proposed a loopy belief propagation variant that is better calibrated on networks with cycles, but at comparable or higher computational expense.

A structurally different approach is the Steiner tree method of Rozenshtein et al. [**?**], which reconstructs the most parsimonious propagation subgraph connecting the observed infected nodes. This model-agnostic topological approach does not require parameter knowledge but does not produce a probabilistic ranking of source candidates. Taken together, these classical methods share a common limitation for the purposes of this thesis: they were designed for static or single-snapshot contact networks and either discard or only partially exploit the temporal ordering of contacts.

3.2 GNN-Based Methods on Static Networks

The application of graph neural networks to source detection was initiated by Dong et al. [**?**], who introduced GCNSI (Graph Convolutional Network for Source Identification), framing the problem as multi-source detection on a single static network. Their key methodological contribution was the use of Label Propagation Algorithm (LPA) outputs as augmented node features, providing the GCN with a soft prior over likely source locations derived from graph structure. The model is trained on simulated spreading outcomes and outputs a binary per-node label. As noted by Sterchi et al. [**?**], the multi-source formulation introduces important differences from the single-source problem studied in this thesis, including a sigmoid output layer that treats nodes independently rather than competing through a softmax normalisation.

Shah et al. [**?**] evaluated both GCN and Graph Attention Network (GAT) variants for single-source detection. Their models take one-hot encoded SIR states as node features and output a softmax distribution over nodes, which is a cleaner formulation for single-source inference. They reported that GNNs outperform DMP on several benchmark networks, providing early evidence that learned message-passing can capture source-relevant structural patterns more effectively than hand-crafted probabilistic approximations.

Subsequent work introduced a range of architectural modifications. The MCGNN of Shu et al. employs a multi-channel architecture in which a line graph representation provides edge-level features alongside the original node graph, aiming to capture higher-order structural information. However, Sterchi et al. [**?**] note that MCGNN incorrectly formulates single-source detection as a multi-label problem with sigmoid activations, undermining its comparability with softmax-based methods. The Infected Graph Convolutional Network (IGCN) proposed by Guo et al. introduces state-dependent aggre-

gation weights, so that the influence of a neighbour on the central node's embedding is modulated by that neighbour's SIR state; this design reflects the intuition that infected and recovered nodes should contribute differently to source inference. The SD-STGCN of Sha et al. extends the input to multiple temporal snapshots of the spreading process, which is a more realistic observation model, but the underlying network is still treated as a static projection.

The systematic review and benchmark of Sterchi, Brack, and Hilfiker [**?**] provides the most comprehensive evaluation of GNN-based source detection methods to date. Their study re-implements representative methods under controlled conditions and evaluates them against classical baselines on a common set of network topologies and epidemic parameters. The key empirical finding is that GNNs substantially and consistently outperform classical methods including DMP, SME, and the Jordan centre. The review also identifies several methodological concerns with the existing literature: claims of model-agnosticism are often made without empirical verification across different spreading models; the SME baseline is frequently absent or configured with an inadequate number of Monte Carlo samples, making comparisons unreliable; and the effect of training set size on GNN performance is understudied. Crucially for this thesis, Sterchi et al. explicitly note that all surveyed GNN methods, with the sole exception of Ru et al.'s Backtracking Network, operate on static projections of the contact network, discarding temporal information entirely.

3.3 GNN-Based Methods on Temporal Networks

The Backtracking Network (BN) of Ru et al. [**?**] represents the current state of the art for GNN-based source detection on temporal contact networks and is the primary reference point for this thesis. Rather than projecting the temporal network to a static graph, the BN retains the full temporal activation structure by attaching to each edge in the aggregated graph a binary vector—referred to as an edge texture—that records at which discrete time steps the contact was active. A kernel-based message-passing operator [**?**, **?**] interleaves edge updates, in which edge hidden states are updated using the hidden state of the source node, with node updates, in which each node aggregates updated messages from all incoming edges. The architecture also incorporates expert knowledge by masking susceptible nodes to zero detection probability prior to the output softmax, reflecting the logical impossibility of a susceptible node being the source. Ru et al. demonstrate that the BN substantially outperforms classical baselines and show robustness to missing links and partially observed node states. The method is described in full technical detail in Chapter 5 of this thesis, as it serves as the temporal GNN baseline against which the De Bruijn GNN and the TEG-DAG-GNN are compared.

Beyond source detection, Rossi et al. proposed Temporal Graph Networks (TGNs) as a general-purpose framework for deep learning on dynamic graphs [**?**]. TGNs maintain a per-node memory state updated by a gated recurrent mechanism as new events arrive, and combine this memory with graph attention over current neighbourhood events. While TGNs have not been applied to source detection, they represent the broader

architectural context within which temporal GNN methods for source detection should be understood: the core challenge is how to encode time-varying edge activity in a way that is informative for node-level classification tasks.

The De Bruijn Graph Neural Network (DBGNN) of Qarkaxhija et al. [**?**] offers a fundamentally different approach to temporal information. Rather than encoding temporal activity as edge features on a static graph, the DBGNN constructs a higher-order De Bruijn graph whose nodes correspond to ordered sequences of contacts, thereby directly representing causal walks and non-Markovian path statistics. This architecture provides an explicit inductive bias toward the causal path structure of temporal spreading processes, which the edge texture approach of the BN can only approximate implicitly. The Temporal Event Graph (TEG) framework of Saramäki et al. [**?**] provides yet another structural representation: it maps a temporal network into a directed acyclic graph whose nodes are contact events and whose directed edges encode time-respecting sequential relationships. Applying the DAG convolutional network of Rey et al. [**?**] to a TEG yields a method that processes the causal structure of the spreading process directly in the spectral domain of an acyclic graph.

3.4 Summary and Research Gap

The existing literature reveals a clear asymmetry: while classical source detection methods have been developed for both static and temporal network settings, GNN-based approaches have been almost entirely confined to static projections. The BN of Ru et al. is the only published GNN method designed specifically for temporal contact networks, and it has not been compared against alternative temporal GNN architectures embodying different inductive biases. No study has systematically evaluated whether representing temporal structure as edge activation patterns (BN), as higher-order causal walks (DBGNN), or as a directed acyclic event graph (TEG-DAG-GNN) yields materially different detection accuracy, or under what conditions each approach is superior. This thesis fills that gap by implementing all three architectures within a unified experimental framework and evaluating them against each other and against static GNN and classical baselines.

4

Methodology: GNNs for Temporal Source Detection

This chapter lays the methodological groundwork for the remainder of the thesis. We begin by formalising the epidemic source detection problem as a probabilistic inference task on temporal contact networks (Section 4.1). We then describe how labelled training data are generated through Monte Carlo simulation of the SIR spreading process (Section 4.2). Finally, we introduce the three distinct representations of temporal contact information used as inputs to the evaluated model architectures (Section 4.3); the architectures themselves are presented in Chapter 5.

4.1 Formalization of the Backtracking Problem

Let $G = (\mathcal{V}, E^t)$ denote a temporal contact network, where $\mathcal{V}$ is a fixed set of N nodes and $E^t = \{(u, v, \tau) : u, v \in \mathcal{V},\ \tau \in \mathbb{N}\}$ is a multiset of timestamped contact events. Each triplet (u, v, τ) records an undirected interaction between nodes u and v at discrete time step τ. The network can equivalently be viewed as a sequence of snapshot graphs $\mathcal{G} = \{G_0, G_1, \ldots, G_T\}$ over the shared node set $\mathcal{V}$, where $G_\tau = (\mathcal{V}, E_\tau)$ and $E_\tau = \{(u, v) : (u, v, \tau) \in E^t\}$.

An SIR spreading process (defined formally in Chapter 2) is initiated by a single source node $v^* \in \mathcal{V}$ at time t_0. The source enters the infectious state I at t_0 while all other nodes are susceptible (S). The process evolves stochastically along the edges of $\mathcal{G}$ until an observation time $t_{\text{end}} = t_0 + T$, at which point each node $v \in \mathcal{V}$ occupies exactly one of the three compartmental states $X_v \in \{S, I, R\}$. The resulting final snapshot is $S_T = \{C_v : v \in \mathcal{V}\}$, where $C_v \in \{0, 1\}^3$ is the one-hot encoding of the state of node v at time t_{end}.

At inference time, two sources of information are available: the final state snapshot S_T and the full temporal contact network $\mathcal{G}$. The exact start time t_0 is not known precisely; it is only assumed to lie within a known interval $[t_{\min}, t_{\max}]$. Crucially, the trajectory of the epidemic between t_0 and t_{end} is unobserved — only the terminal state is accessible. This partial observability is the defining characteristic of the backtracking problem and its primary source of difficulty.

The source detection objective is to infer a probability distribution $\hat{p} \in \Delta^{N-1}$ over all nodes, where $\hat{p}_i$ approximates the posterior probability $P(v^* = v_i \mid S_T, \mathcal{G})$. The

node $\hat{v}^* = \arg\max_i \hat{p}_i$ is returned as the predicted source. Formally, we seek to learn a parametric function

$$f_\theta : (G_a,\, \mathbf{C}) \;\longrightarrow\; \hat{p} \in \Delta^{N-1},$$

where G_a is the graph structure consumed by the GNN and $\mathbf{C} \in \{0,1\}^{N\times 3}$ is the node feature matrix whose i-th row is C_{v_i}. The precise form of G_a depends on the temporal representation used and varies across the evaluated models, as discussed in Section 4.3.

The problem is subject to a single-source assumption: exactly one node v^* initiates the outbreak. Under this assumption, a powerful structural constraint holds: any node that is still susceptible at observation time t_{end} cannot be the source, since the source was infectious at t_0 and therefore can never return to the susceptible state. This *expert knowledge mask* eliminates all susceptible nodes from the candidate set and is applied by all GNN architectures evaluated in this thesis (see the `BacktrackingNetwork` implementation in `gnn/backtracking_network.py`). The detection problem is made hard by two compounding factors. First, the SIR spreading process is inherently stochastic: even when the source and the network are held fixed, repeated realisations produce substantially different final snapshots, meaning the mapping from v^* to S_T is many-to-many. Second, given only the final snapshot, many different source nodes may have produced equally plausible outcomes, leading to a multimodal and diffuse posterior $P(v^* \mid S_T, \mathcal{G})$.

4.2 Data Generation via Monte Carlo Simulations

Supervised training of a GNN for source detection requires a large collection of labelled instances, each pairing a final SIR snapshot S_T with the corresponding ground-truth source v^*. Since labelled epidemiological data of this form are not available at scale in real-world settings, all training data are generated synthetically by simulating the SIR process on the target temporal network.

For each training instance, a source node v^* is selected uniformly at random from those nodes that are active — i.e., present in at least one contact — at a randomly sampled start time $t_0 \in [t_{\min}, t_{\max}]$. The SIR process is then run stochastically on $\mathcal{G}$ from t_0 to t_{end}, producing a final state vector S_T. The pair (S_T, v^*) constitutes one labelled training instance. Because the spreading process is stochastic, multiple independent realisations from the same source v^* are simulated, exposing the model to the natural variance in outbreak trajectories from a given starting point.

Two distinct sets of simulations are generated per source node. *Ground-truth runs* consist of a large number M_{gt} of independent realisations per source. These are used exclusively at evaluation time: by averaging predictions over many realisations, the variance in model performance due to stochasticity in S_T is reduced, yielding a stable estimate of source detectability from each possible starting node. *Monte Carlo training runs* consist of a smaller number M_{mc} of independent realisations per source, drawn from the same distribution. These form the actual training set: a total of $N \times M_{\text{mc}}$ instances, one for each (source, realisation) pair. The separation between ground-truth and training runs ensures that model performance is evaluated on realisations that were never seen

during training, providing an unbiased estimate of generalisation. The resulting data are stored as flat binary arrays of type `int8`, with ground-truth and Monte Carlo states for each compartment (S, I, R) stored in separate files and reshaped to tensors of shape (N, M, N) at load time, as implemented in `setup/data_loader.py`.

Not every simulated outbreak provides useful training signal. Outbreaks that fail to spread beyond the immediate neighbourhood of the source, or that extinguish after only a single infection event, offer little information about the network's contact structure and may bias the model towards trivially detectable cases. Simulations are therefore filtered by minimum outbreak size: any realisation in which the number of infected or recovered nodes at t_{end} falls below a configurable threshold is discarded. This ensures that the training distribution consists of outbreaks that have meaningfully propagated through the network. An additional *possible-sources mask* is computed for each ground-truth run: a node v is flagged as a feasible source candidate for run r if and only if it is non-susceptible in that run, i.e., $C_v \neq S$. This binary mask, of shape (N, M_{gt}, N), is used to suppress logically impossible source assignments during evaluation.

Data generation is implemented in the `tsir/` module of this project, which wraps Holme's fast C-based temporal SIR simulation engine [**?**]. The C implementation uses an event-driven algorithm and is substantially faster than equivalent pure-Python realisations, enabling the generation of the large simulation volumes required for supervised training. The C code is invoked from Python via a subprocess interface, and all output is written to disk as binary arrays. Each complete dataset — comprising ground-truth runs, Monte Carlo runs, the temporal network graph, and the possible-sources mask — is logged as a versioned artifact to the Weights and Biases (*W&B*) experiment tracking platform. Downstream training and evaluation runs reference the artifact by name and version, guaranteeing that every model is trained and evaluated on identical data regardless of when or where the run is executed.

4.3 Representation of Temporal Information

The three GNN architectures evaluated in this thesis operate on the same underlying simulation data but differ fundamentally in how the temporal contact structure of $\mathcal{G}$ is encoded into a graph that can be processed by a GNN. This section describes the three representations in turn. Each is constructed from the temporal graph $\mathcal{G}$ with its associated edge timestamp lists; the corresponding construction code resides in `gnn/graph_builder.py`. The choice of representation determines which temporal patterns are preserved, which are discarded, and how the resulting graph interacts with the message-passing operations of the corresponding model architecture.

4.3.1 Static Projection (Baseline)

The simplest representation of a temporal contact network is obtained by collapsing the full temporal sequence $\mathcal{G} = \{G_0, G_1, \ldots, G_T\}$ into a single static graph $G_a = (\mathcal{V}, E_a)$,

where the aggregated edge set is

$$E_a = \bigcup_{\tau=0}^{T} E_\tau.$$

Every node pair that had at least one contact at any time step appears as a single undirected edge in G_a. To retain a measure of contact frequency, edge weights are assigned as $w_{uv} = |\{\tau : (u, v) \in E_\tau\}|$, counting the number of distinct time steps at which the contact was active. When edge weights are used, they are normalised to the unit interval $[0, 1]$ by dividing by the maximum observed count. This representation retains degree structure and contact frequency information but discards all temporal ordering: it is impossible to determine from G_a alone whether a given edge was active early or late in the observation window. The static projection is consumed by the `StaticGNN` baseline, implemented in `gnn/static_gnn.py`, and also provides the structural backbone (without the temporal edge attributes) for the Backtracking Network.

4.3.2 Temporal Discretization (Timeslices)

The timeslice representation preserves the full temporal ordering of contacts by maintaining a separate edge set for each time step, yielding the sequence $\{E_0, E_1, \ldots, E_T\}$. In practice, many time steps may contain few or no contacts, and processing one GNN layer per original time step is computationally prohibitive for long observation windows. Consecutive time steps are therefore grouped into bins of width Δt, reducing the effective number of slices to $\lfloor T/\Delta t \rfloor$. All contacts within a window $[\tau\Delta t, (\tau + 1)\Delta t)$ are merged into a single snapshot edge set $E_\tau^{\Delta t}$. The parameter Δt is controlled by the `group_by_time` hyperparameter in the experimental configuration. As $\Delta t \to T$, the timeslice representation converges to the static projection, providing a natural interpolation between fully temporal and fully aggregated representations.

The timeslice sequence $\{E_0^{\Delta t}, \ldots, E_{\lfloor T/\Delta t \rfloor}^{\Delta t}\}$ is consumed by the `TemporalGNN`, implemented in `gnn/temporal_gnn.py`. This architecture applies one dedicated graph convolutional layer (specifically, a GraphSAGE convolution [**?**]) per time slice, processing the slices in reverse chronological order — from t_{end} back towards t_0 — in order to propagate information backwards along the direction of potential infection pathways. The reversed traversal is a deliberate design choice motivated by the backtracking objective: features observed at t_{end} are iteratively propagated to earlier contact neighbourhoods to identify which nodes could plausibly have been active sources.

4.3.3 Edge Feature Encoding for Timestamps

The edge texture representation, introduced by Ru et al. [**?**], encodes the complete temporal activation history of each contact as a fixed-length binary vector attached to the corresponding edge in the aggregated static graph G_a. For each edge $(u, v) \in E_a$, the activation vector $X_{(u,v)} \in \{0, 1\}^T$ is defined entry-wise as

$$\big[X_{(u,v)}\big]_\tau = \mathbf{1}\big[(u, v) \in E_\tau\big], \quad \tau = 0, 1, \ldots, T - 1.$$

The full set of activation vectors forms an edge feature matrix $X_a \in \{0,1\}^{|E_a| \times T}$, where $|E_a|$ counts directed edge pairs (both (u,v) and (v,u) are included with the same activation pattern, reflecting the undirected nature of the contacts). This representation preserves the exact temporal pattern of every contact — including the sequential order of activations, the gaps between active periods, and the total contact duration — within a compact fixed-size descriptor that does not depend on the total number of unique time steps visited. The graph G_a together with the matrix X_a constitutes the full input to the `BacktrackingNetwork` [**?**], implemented in `gnn/backtracking_network.py`. In that architecture, X_a is projected into a hidden embedding space via a learned linear layer before the convolutional message-passing layers, allowing the model to identify temporal patterns in the contact history that are informative for attributing the observed final state S_T to a particular source node.

5

Model Architectures and Implementation

This chapter describes the four GNN architectures evaluated in this thesis. The static projection baseline operates on the time-aggregated contact graph and serves as the lower bound against which temporal methods are compared. The three temporal architectures—the Backtracking Network, the De Bruijn GNN, and the DAG-GNN on Temporal Event Graphs—each encode the temporal contact history under a different inductive bias, as motivated in Chapter 3.

5.1 Evaluated GNN Models

5.1.1 Static Projection (Baseline)

The static projection baseline collapses the temporal contact network into a single static graph and applies a standard GNN for source detection. This model intentionally discards all temporal information and serves as the primary reference point for quantifying the benefit of temporal representations.

Graph Construction

Given the temporal contact network $\mathcal{G} = \{G_0, G_1, \ldots, G_T\}$, the aggregated static graph $G_a = (\mathcal{V}, E_a)$ is obtained by taking the union of all snapshot edge sets:

$$E_a = \bigcup_{t=0}^{T} E_t.$$

Each edge $(u, v) \in E_a$ may optionally carry a contact-count weight $w_{uv} = |\{t : (u, v) \in E_t\}|$, reflecting the number of time steps at which the contact was active. Node features are the one-hot encoded final SIR states $\mathbf{C} \in \{0, 1\}^{N \times 3}$, identical to those used by the Backtracking Network.

GNN Architecture

The static GNN applies L graph convolutional layers to the aggregated graph. Following Sterchi et al. [**?**], a standard GCN layer update is used:

$$\mathbf{h}_v^{(l)} = \mathrm{ReLU}\left(\mathbf{W}^{(l)} \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{\mathbf{h}_u^{(l-1)}}{\sqrt{\tilde{d}_v \, \tilde{d}_u}}\right), \qquad l = 1, \ldots, L,$$

where $\tilde{d}_v$ is the degree of node v augmented by one to account for the self-loop. After L layers, a final linear projection maps each node embedding to a scalar score, and the expert knowledge mask suppresses susceptible nodes before the softmax, as described in Section 5.1.3. By ignoring temporal ordering, the static projection provides a controlled lower bound on source detection performance and isolates the marginal contribution of temporal information in the architectures that follow.

5.1.2 De Bruijn Graph Neural Networks (DBGNN)

Standard GNNs operate under the Markovian assumption, whereby node influence depends only on immediate neighbours. The De Bruijn Graph Neural Network (DBGNN) proposed by Qarkaxhija et al. [**?**] overcomes this limitation by lifting the temporal network into a higher-order state space that explicitly represents causal walks and non-Markovian dynamics.

Causal Walks and Higher-Order Graph Construction

A sequence of timestamped contacts $((v_0, v_1, t_1), (v_1, v_2, t_2), \ldots, (v_{k-1}, v_k, t_k))$ is a *causal walk* of length k if contacts occur in strictly increasing order, $t_1 < t_2 < \cdots < t_k$, and consecutive contacts occur within a maximum inter-event time δ, i.e. $t_{i+1} - t_i \leq \delta$ for all i. The set of all causal walks characterises which transmission chains are temporally feasible given the contact ordering.

The k-th order De Bruijn graph $G^{(k)} = (V^{(k)}, E^{(k)})$ is constructed from all causal walks of length $k-1$. Each node $u \in V^{(k)}$ represents one such walk, and $\pi(u) = \{v_0, \ldots, v_{k-1}\}$ denotes its projection onto constituent vertices. A directed edge exists from $u \in V^{(k)}$ to $w \in V^{(k)}$ if and only if

$$(v_1, \ldots, v_{k-1}) = (w_0, \ldots, w_{k-2}) \quad \text{and} \quad t_{u,k-1} < t_{w,k-1},$$

where $t_{u,k-1}$ denotes the timestamp of the final contact in walk u. The initial feature of node u is

$$\mathbf{h}_u^{(0)} = \mathrm{MLP}\big(\mathbf{x}_{v_0}, \ldots, \mathbf{x}_{v_{k-1}}\big)\,,$$

where $\mathbf{x}_{v_i} \in \{0,1\}^3$ is the one-hot SIR state of node v_i at the final observation time T.

Message Passing

The hidden state of node $u \in V^{(k)}$ at layer $l+1$ is updated via a GraphSAGE-style mean aggregation over its causal-walk neighbours:

$$\mathbf{h}_u^{(l+1)} = \sigma\Big(\mathbf{W}^{(l)} \cdot \mathrm{MEAN}\Big(\{\mathbf{h}_u^{(l)}\} \cup \{\mathbf{h}_w^{(l)} : w \in \mathcal{N}_{G^{(k)}}(u)\}\Big)\Big)\,,$$

where σ denotes a non-linear activation function and $\mathbf{W}^{(l)} \in \mathbb{R}^{d_l \times d_{l-1}}$ is a learnable weight matrix.

Readout and Source Prediction

The higher-order embeddings must be mapped back to original nodes $v \in V$ to produce a distribution over candidate sources. This is achieved via scaled dot-product attention pooling over all walks containing v:

$$\mathbf{h}_v^{\text{final}} = \sum_{u \in V^{(k)},\, v \in \pi(u)} \alpha_u \, \mathbf{h}_u^{(L)}, \qquad \alpha_u = \frac{\exp\left(\dfrac{\mathbf{q}^\top \mathbf{h}_u^{(L)}}{\sqrt{d_L}}\right)}{\displaystyle\sum_{u' \in V^{(k)},\, v \in \pi(u')} \exp\left(\dfrac{\mathbf{q}^\top \mathbf{h}_{u'}^{(L)}}{\sqrt{d_L}}\right)},$$

where $\mathbf{q} \in \mathbb{R}^{d_L}$ is a global learnable query vector and d_L is the hidden dimension at the final layer. A linear layer followed by the susceptibility mask and softmax produces the output source distribution.

5.1.3 Kernel-based GNN / Backtracking Network (BN)

The Backtracking Network (BN) of Ru et al. [**?**] is the first graph neural network method designed specifically for epidemic source detection on temporal contact networks. It is the primary temporal GNN baseline in this thesis and the architecture against which the De Bruijn GNN and the TEG-DAG-GNN are evaluated.

Problem Formulation

The temporal contact network is represented as a sequence of static snapshot graphs $\mathcal{G} = \{G_0, G_1, \ldots, G_T\}$ sharing a common node set $\mathcal{V}$ of size N. Each snapshot $G_t = (\mathcal{V}, E_t)$ records which pairs of nodes were in contact at discrete time step t. The epidemic spreading process follows the SIR model: at each time step, an infected node i transmits the pathogen to each susceptible neighbour $j \in \mathcal{N}(i)$ with probability p, and independently recovers with probability q.

In practice the exact start time of the outbreak is not known. The posited temporal network $\tilde{\mathcal{G}}$ encompasses a window $[t_{\min}, t_{\max}]$ that is estimated to contain the true index event, but the precise start time $t_0 \in [t_{\min}, t_{\max}]$ is treated as a latent variable. The source detection task is then: given the observed final state S_T and the posited network $\tilde{\mathcal{G}}$, infer the initial source node S_0 (patient zero). Ru et al. express this as the learned mapping

$$S_0 = \mathbf{BN}(S_T,\ \tilde{\mathcal{G}}).$$

Encoding Available Information

Two sources of information are provided to the model as input features.

Node features—final state S_T. Each node's SIR state at time T is one-hot encoded as a three-dimensional feature vector $C_i \in \{0, 1\}^3$, with susceptible nodes represented

as $[1, 0, 0]$, infectious nodes as $[0, 1, 0]$, and recovered nodes as $[0, 0, 1]$. The full node feature matrix is $\mathbf{C} \in \{0, 1\}^{N \times 3}$.

Edge features—binary activation patterns (edge textures). The temporal structure of the contact network is encoded by first constructing the aggregated static graph $\tilde{G}_a = (\mathcal{V}, E_a)$, where $E_a = \bigcup_{t=0}^{T} E_t$ contains every edge that appears in at least one snapshot. Each edge $e \in E_a$ is assigned a binary activation vector $X_e \in \{0, 1\}^T$, where entry t equals one if and only if the edge was active in snapshot G_t. The full edge feature matrix is $X_a = \{X_e \mid e \in E_a\} \in \{0, 1\}^{|E_a| \times T}$. This representation, which Ru et al. call an *edge texture*, compresses the full temporal contact history into a fixed-length binary descriptor attached to each edge in the aggregated graph.

Architecture

The BN processes the input through three sequential components: initial feature projections, a stack of L kernel-based convolutional layers, and a final projection to scalar per-node scores.

Initial projections. The raw node and edge features are projected into a common D-dimensional hidden space. The node projection $p^v : \mathbb{R}^3 \to \mathbb{R}^D$ and edge projection $p^e : \mathbb{R}^T \to \mathbb{R}^D$ are each a single linear layer followed by a ReLU activation:

$$h_i^0 = p^v(C_i), \qquad g_{(i,j)}^0 = p^e\big(X_{(i,j)}\big)\,,$$

where $h_i^0 \in \mathbb{R}^D$ and $g_{(i,j)}^0 \in \mathbb{R}^D$ denote the initial hidden representations of node i and edge (i, j), respectively.

Kernel-based convolutional layers. Each of the L layers applies a kernel-based convolutional operator [**?**, **?**] consisting of two interdependent update equations. The edge update reads the current edge hidden state together with the hidden state of the source node and produces a new edge representation:

$$g_{(i,j)}^{l+1} = f^e\Big(\Big[g_{(i,j)}^l,\ h_i^l\Big]\Big)\,,$$

where $[\cdot, \cdot]$ denotes concatenation and f^e is a fully connected layer with ReLU activation. The node update then aggregates the newly computed incoming edge messages and applies a self-transformation:

$$h_i^{l+1} = \mathrm{ReLU}\left(f^v\Big(h_i^l\Big) + \sum_{j \in \mathcal{N}(i)} g_{(j,i)}^{l+1}\right),$$

where f^v is an independent fully connected layer with ReLU activation. The design couples edge and node representations across layers: edge states are informed by the source node's embedding, and node states are informed by all incoming edge messages.

Final projection. After L convolutional layers, the node hidden state $h_i^L \in \mathbb{R}^D$ is projected to a scalar score $h_i \in \mathbb{R}$ via a final linear layer.

Expert Knowledge and Output Distribution

A key design feature of the BN is the injection of domain knowledge as a hard constraint on the output distribution. A node that is susceptible at the final observation time T cannot logically be the epidemic source: the source must have been infected at $t_0 \leq T$, and under SIR dynamics a susceptible node has never been infected. The BN enforces this constraint by setting the output logit of every susceptible node to negative infinity before the softmax:

$$h_s \leftarrow -\infty \quad \forall\, s \in \mathcal{V} \text{ such that } C_s = [1, 0, 0].$$

The per-node log-probabilities are then computed as

$$\log \hat{p}_i = \log \frac{\exp(h_i)}{\sum_{j=1}^{N} \exp(h_j)},$$

which assigns probability zero to all susceptible nodes without modifying any network weights.

Training Objective and Procedure

The model is trained by minimising the cross-entropy between the predicted log-probabilities and the one-hot label indicating the true source:

$$\mathcal{L} = -\sum_{i=1}^{N} y_i \log \hat{p}_i,$$

where $y_i = 1$ if node i is the true patient zero and $y_i = 0$ otherwise. The loss is minimised using the Adam optimiser [**?**] with a learning rate of 10^{-3} and a batch size of 128. Ru et al. report that optimal performance is achieved at $L = 5$ convolutional layers; performance saturates or degrades at higher layer counts, consistent with the over-smoothing phenomenon in GNNs.

Training samples are generated by Monte Carlo simulation of the SIR spreading process on the temporal network. For each sample, a start time t_0 is drawn uniformly at random from $[t_{\min}, t_{\max}]$, a patient zero is selected from the set of nodes active at t_0, and the SIR process is simulated until t_{end}. This procedure is implemented using an adapted version of Holme's fast C-based temporal SIR simulation engine [**?**], wrapped in the project's `tsir/` module, with data logged as versioned artefacts via Weights and Biases.

Robustness to Missing Information

Ru et al. evaluate the BN under two degraded-information scenarios. In the first, a random fraction of edges is removed from each snapshot, simulating incomplete knowledge of the contact network structure. In the second, only a random fraction of node states is observed in the final snapshot S_T, with the remaining nodes assigned an unknown

state. The BN maintains a Top-5 accuracy of approximately 0.7 even when 80% of final node states are unobserved, suggesting that the edge texture representation provides a strong structural prior. Performance degrades more sharply when the contact network structure itself is incomplete, as the edge texture is the primary carrier of temporal information in the model.

Critical Assessment

Sterchi et al. [**?**] raise an important concern about the evaluation in Ru et al.: the comparison with SME uses only 200 Monte Carlo simulations per candidate source, a strongly sub-optimal configuration that likely underestimates SME's true accuracy. The reported advantage of the BN over SME may therefore be overstated. This thesis addresses this limitation by evaluating all baselines, including SME, with a substantially larger number of simulations. At a deeper architectural level, the edge texture representation is a lossy encoding of the temporal contact history: it encodes temporal ordering implicitly through the binary pattern, but does not explicitly model causal walk structure. This is precisely the limitation that the De Bruijn GNN and the TEG-DAG-GNN are designed to address.

5.1.4 Directed Acyclic Graph GNN (DAG-GNN) on Temporal Event Graphs

The DAG-GNN combines the Temporal Event Graph (TEG) representation of Saramäki et al. [**?**] with the Directed Acyclic Graph Convolutional Network (DCN) of Rey et al. [**?**]. Rather than attaching temporal information as edge features on a static graph (as the BN does), this approach transforms the temporal network into a new graph whose structure directly encodes causal relationships between contact events.

Temporal Event Graph Construction

Given the temporal contact network $\mathcal{G} = \{G_0, \ldots, G_T\}$, the TEG is a directed graph $\mathcal{H} = (\mathcal{V}_\mathcal{H}, \mathcal{E}_\mathcal{H})$ in which each node $e_i \in \mathcal{V}_\mathcal{H}$ corresponds to a contact event $(u_i, v_i, t_i) \in \bigcup_t E_t$. A directed edge exists from event e_j to event e_i if e_j is the most recent contact involving the node u_i before the event e_i occurs. Formally, the predecessor set of event $e_i = (u_i, v_i, t_i)$ is

$$\mathrm{Pre}(e_i) = \big\{e_j = (w,\, u_i,\, t_j) \in \mathcal{V}_\mathcal{H} : t_j < t_i \;\wedge\; \nexists\, e_k = (z,\, u_i,\, t_k) \text{ s.t. } t_j < t_k < t_i\big\}.$$

This construction ensures that directed edges in the TEG trace the immediate temporal predecessors of each contact, producing a DAG whose topological order respects the causal structure of the spreading process. The TEG has $|\bigcup_t E_t|$ nodes and encodes the full sequence of inter-event dependencies without any loss of temporal ordering.

Feature Encoding

Each event node $e_i \in \mathcal{V}_{\mathcal{H}}$ is initialised with a feature vector concatenating the one-hot SIR states of its two constituent nodes at the observation time T and a sinusoidal temporal encoding of the event timestamp:

$$\mathbf{x}_{e_i} = \Big[\mathbf{C}_{u_i} \,\|\, \mathbf{C}_{v_i} \,\|\, \mathrm{TE}(t_i)\Big],$$

where $\mathbf{C}_{u_i}, \mathbf{C}_{v_i} \in \{0,1\}^3$ are the one-hot states and $\mathrm{TE}(t_i)$ is the sinusoidal positional encoding

$$\mathrm{TE}(t_i) = \Big[\sin\Big(\tfrac{t_i}{10000^{2k/d}}\Big),\ \cos\Big(\tfrac{t_i}{10000^{2k/d}}\Big)\Big]_{k=1}^{d/2}.$$

This encoding allows the model to distinguish events not only by the SIR states of the involved nodes but also by their absolute position in the temporal sequence.

DAG Convolutional Layers

The DAG-GNN processes event nodes in topological order, propagating information forward through the causal event chain. At each layer l, the hidden state of event e_i is updated by aggregating messages from its predecessors:

$$\mathbf{z}_{e_i}^{(l+1)} = \mathrm{MLP}_{\mathrm{upd}}\left(\left[\mathbf{x}_{e_i} \,\Big\|\, \sum_{e_j \in \mathrm{Pre}(e_i)} \mathrm{MLP}_{\mathrm{msg}}\Big(\Big[\mathbf{z}_{e_j}^{(l)} \,\|\, e^{-\lambda\,\Delta t_{ji}}\Big]\Big)\right]\right),$$

where $\Delta t_{ji} = t_i - t_j$ is the inter-event temporal gap and λ is a learnable non-negative decay parameter that down-weights temporally distant predecessors. Processing in topological order is critical here: each event's hidden state is fully determined before it is used as a message for subsequent events, a property guaranteed by the acyclic structure of the TEG. This is the key distinction from standard GNNs, in which circular message passing across iterations can mix information from temporally incompatible contact sequences.

Readout and Source Prediction

After L layers, the final event embeddings $\{\mathbf{z}_{e_i}^{(L)}\}$ are aggregated back to original network nodes $v \in \mathcal{V}$ by mean-pooling over all events in which v participated:

$$\mathbf{h}_v = \frac{1}{|\{e_i : v \in \{u_i, v_i\}\}|} \sum_{\substack{e_i \in \mathcal{V}_{\mathcal{H}} \\ v \in \{u_i, v_i\}}} \mathbf{z}_{e_i}^{(L)}.$$

A final linear layer maps $\mathbf{h}_v \in \mathbb{R}^D$ to a scalar score, the susceptibility mask suppresses impossible sources, and the softmax produces the output distribution over candidate source nodes.

5.2 Hyperparameter Optimisation and Training Setup

All four model architectures are trained within a common experimental framework to ensure comparability. Models are optimised using the Adam optimiser [**?**] with a fixed learning rate of 10^{-3} and weight decay 10^{-4}. Training proceeds for a maximum of 500 epochs with early stopping: if the validation loss does not improve for 30 consecutive epochs, training is terminated and the checkpoint with the lowest validation loss is restored. All models use a batch size of 128 training samples drawn uniformly at random from the pool of Monte Carlo simulations. The training and validation sets are split 80/20 stratified by source node, ensuring that each source is represented in both partitions.

Model-specific hyperparameters — including the number of layers L, hidden dimension D, and dropout rate — are tuned per architecture using a grid search over a small held-out validation set derived from the training data. The layer count for the Backtracking Network follows Ru et al. and is fixed at $L = 5$. The De Bruijn GNN order k is set to 2 for all experiments; the maximum inter-event time δ is set to the 90th percentile of inter-event gaps in the training network. The DAG-GNN uses $L = 4$ layers with a hidden dimension of $D = 64$. The static GNN baseline uses the same hidden dimension and layer count as the BN for a fair comparison. All hyperparameters and training configurations are recorded in YAML files under `exp/` and logged automatically to Weights and Biases at the start of each training run.

6

Experimental Evaluation

This chapter describes the experimental framework used to evaluate the four GNN architectures introduced in Chapter 5 against a set of classical heuristic baselines. We specify the datasets employed, define the evaluation metrics, and outline the analyses that will be performed once training runs are complete. The experiments are designed to test the three research hypotheses stated in Chapter 1: that temporal GNN architectures outperform static projections on structured networks (H1), that this advantage diminishes as the observation window grows (H2), and that temporal models maintain robustness in late-stage outbreaks where the aggregated contact history is dense (H3).

6.1 Datasets

6.1.1 Synthetic Graphs

We generate two families of synthetic temporal contact networks to provide controlled experimental conditions. The first family follows the Erdős–Rényi random graph model $G(N, p)$, with node counts $N \in \{50, 100, 200, 500\}$ and edge probability $p = 2/N$, placing all instances in the sparse regime where the mean degree remains approximately two regardless of N. Temporal structure is imposed by treating each undirected static edge as a potential contact: at each of T discrete time steps the edge is independently activated with probability p_{act}, yielding a sequence of contact matrices that respects the sparse backbone of the random graph. This activation model ensures that the number of temporal contacts scales linearly with T while keeping the instantaneous network sparse, a regime broadly representative of real face-to-face proximity data.

The second family uses the Barabási–Albert preferential attachment model with attachment parameter $m \in \{1, 2, 3\}$, producing scale-free degree distributions whose heterogeneity is known to accelerate epidemic spreading and to concentrate infection probability near high-degree hubs. The same independent temporal activation model is applied. Across both families, SIR simulations are parameterised with infection rate $\beta \in \{0.1, 0.2, 0.3\}$, recovery rate $\mu \in \{0.05, 0.1, 0.2\}$, and observation window $T \in \{10, 20, 40\}$ time steps, yielding a full factorial grid of spreading conditions. For each combination of network instance and parameter setting, 5000 Monte Carlo simulations are generated per candidate source node for training, and 1000 ground-truth simulations per source are reserved for evaluation, providing sufficiently stable empirical accuracy estimates.

6.1.2 Real-world Contact Networks

To assess generalisation to realistic epidemic scenarios, we evaluate all models on five empirical temporal networks drawn from the literature on infectious disease spreading. The Sexual contact network (Rocha et al. 2011) encodes 9319 timestamped interactions between 5762 individuals and represents an undirected face-to-face proximity setting of direct epidemiological relevance. The Hospital network (Isella et al. 2011) comprises 1139 contacts among 75 nodes recorded by RFID sensors in a French hospital ward, providing a small, dense, undirected graph in which SIR dynamics are tightly constrained by the ward topology. The European email network (Leskovec et al. 2007) contains 2113 directed messages exchanged among 625 members of a European research institution, introducing directional flow and thus non-trivial causal path structure. The Bitcoin trust network (Kumar et al. 2016) records 4053 directed trust ratings among 1302 users of a cryptocurrency platform, while the Messages network (Panzarasa et al. 2009) contains 8764 directed private messages exchanged between 1522 students of an online community. The two directed networks introduce asymmetric spreading dynamics that are absent from the undirected synthetic benchmarks.

SIR simulations on all real-world networks follow the protocol of Ru et al. [**?**]: spreading parameters are drawn from the set $(\bar{p}, \bar{q}) \in \{(0.3, 0.01), (0.8, 0.05)\}$, where $\bar{p}$ and $\bar{q}$ denote per-contact infection and recovery probabilities respectively. The epidemic start time is drawn uniformly at random from the interval $[t_{\min}, t_{\max}]$ of recorded contact times, ensuring that models cannot exploit a fixed outbreak onset. The face-to-face networks—Sexual contact and Hospital—are considered the most epidemiologically realistic benchmarks, as the undirected and temporally dense contact structure closely mirrors the assumptions underlying the SIR model.

6.2 Evaluation Metrics

6.2.1 Top-k Accuracy

The primary evaluation metric is Top-k accuracy, which measures the fraction of test instances for which the true source node appears among the k highest-ranked candidates according to the model's predicted probability distribution. Formally, let Q denote the set of test instances, v_q^* the true source for instance q, and $\mathrm{topk}(\hat{p}_q)$ the set of k nodes receiving the highest predicted probability $\hat{p}_q$ under the model. Then

$$\text{Top-k} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\left[v_q^* \in \mathrm{topk}(\hat{p}_q)\right].$$

We report Top-1, Top-3, and Top-5 accuracy throughout. Top-1 accuracy is the strictest criterion and directly measures exact source identification. Top-3 and Top-5 accuracies are motivated by the practical demands of contact tracing: in a real outbreak investigation, public health authorities can typically follow up on a short ranked list of

suspects rather than requiring a single deterministic answer. Reporting across all three thresholds thus bridges the gap between algorithmic precision and operational utility.

6.2.2 Average Error Distance and Reciprocal Rank

Mean Reciprocal Rank (MRR) provides a continuous summary of the model's global ranking quality. Let $\mathrm{rank}(v_q^*)$ denote the position of the true source in the list of nodes sorted by descending predicted probability. Then

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\mathrm{rank}(v_q^*)}.$$

MRR penalises models that place the true source far down the ranked list more severely than Top-k accuracy, because the harmonic weighting by rank discounts partial success rapidly. A model that consistently ranks the true source second or third will have an MRR notably below one even if its Top-3 accuracy is high; MRR is therefore sensitive to systematic underconfidence near the top of the distribution.

We complement MRR with a hop-distance metric that acknowledges the practical value of a near-miss prediction. Define $\hat{v}_q^* = \arg\max_v \hat{p}_q(v)$ as the top-ranked predicted source, and let $d(\cdot,\cdot)$ denote shortest-path distance on the aggregated static graph G_a obtained by collapsing all temporal contacts. The Hop-1 score is then

$$\text{Hop-1} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\big[d\big(v_q^*,\, \hat{v}_q^*\big) \le 1\big]\,.$$

A prediction is counted as a practical success whenever the estimated source is the true source or one of its direct neighbours, since the true source is then contained in the immediate contact set of the predicted node and is thus recoverable by a single tracing step.

6.2.3 Resistance Score for Uncertainty Quantification

Beyond point-estimate accuracy, we evaluate the stability of each model's inferences when the assumed spreading parameters deviate from those used at training time. Let $\mathcal{P}$ be a held-out set of perturbed parameter pairs (β', μ') that differ from the training parameters (β, μ). For each perturbation, we compute the ranked ordering of nodes produced by the model under the training parameters and under the perturbed parameters, and measure their agreement via Kendall's τ rank correlation. The Resistance Score is defined as

$$\mathcal{R} = \frac{1}{|\mathcal{P}|} \sum_{(\beta',\mu') \in \mathcal{P}} \text{Kendall-}\tau\big(\mathrm{rank}_{\beta,\mu},\, \mathrm{rank}_{\beta',\mu'}\big)\,.$$

Kendall's $\tau \in [-1, 1]$ equals one when both rankings are identical and zero when they are uncorrelated. A high value of $\mathcal{R}$ indicates that the model's posterior over

source nodes is robust to misspecification of the epidemic parameters—a property of considerable practical importance, since β and μ are rarely known precisely in a real outbreak. Models that rely heavily on fine-grained parameter-dependent features are expected to yield low $\mathcal{R}$, whereas architectures that capture structural properties of the observed spreading pattern are expected to be more resilient.

6.3 Results and Analysis

6.3.1 Performance Comparison: GNNs vs. Classical Heuristics

The primary experimental comparison will evaluate all four GNN architectures—the Static GNN baseline, the Backtracking Network (BN) of Ru et al. [**?**], the De Bruijn GNN (DBGNN) of Qarkaxhija et al., and the DAG-GNN on Temporal Event Graphs—against five classical heuristics: uniform assignment, random ranking, degree centrality, Jordan centre, and the Soft Margin Estimator (SME). Results will be reported in terms of Top-1, Top-3, Top-5, MRR, and Hop-1 for each dataset and parameter combination. Following the benchmark findings of Sterchi et al. [**?**], we expect all four GNN architectures to substantially outperform the non-learning baselines, with the margin being largest on the structured real-world networks where heuristics based solely on node centrality are poorly calibrated to the actual spreading dynamics.

Among the GNN architectures, we expect the static projection baseline to form a lower bound on GNN performance, with temporal architectures providing incremental gains whose magnitude depends on the degree to which temporal ordering disambiguates the set of plausible sources. The BN's explicit edge texture representation encodes the full temporal contact sequence for each incident edge and provides a strong structural prior that we expect to translate into a consistent advantage over the static baseline across most settings (H1). The DBGNN and DAG-GNN operate on higher-order causal representations and may surpass the BN when the network topology produces rich causal walk structures that edge textures alone do not fully capture. The relative ordering of these two temporal architectures will be examined in detail across network families to determine whether the causal graph representation or the kernel-based texture approach extracts more predictive information for source localisation.

6.3.2 Impact of Network Density and Observation Time T

A dedicated ablation will vary N, the Erdős–Rényi density p, and the observation window T independently while holding the remaining parameters fixed. As the observation window T increases, a larger fraction of the network's static edges will have been activated at least once, causing the aggregated contact graph G_a to approach the full static backbone and eroding the informational advantage of temporal ordering. We expect this effect to produce a systematic convergence of temporal and static GNN performance as T grows, directly testing hypothesis H2. Specifically, for large T the Static GNN's loss of temporal information becomes less consequential because most causal paths are realised

within the observation window, leaving fewer alternative paths to be pruned by temporal constraints.

The effect of network density is expected to interact with the choice of architecture. In sparse Erdős–Rényi graphs, the contact network has few cycles and temporal ordering imposes relatively weak pruning of the causal path set; the advantage of causal walk representations is therefore anticipated to be modest. In denser networks or in scale-free graphs where high-degree hubs create many alternative short paths, the causal walk pruning performed by the DBGNN should become increasingly valuable relative to the static projection, since a larger fraction of static paths are not temporally reachable. Similarly, larger N with fixed mean degree $p = 2/N$ will be used to assess the scaling behaviour of each architecture, since the cost of constructing the De Bruijn graph and the Temporal Event Graph both grow with the number of observed contacts.

6.3.3 Robustness to Missing Information (Links/Node States)

Following the experimental protocol of Ru et al. [**?**], we will evaluate each model under two types of partial observation. In the first setting, a fraction $\rho_e \in \{0.1, 0.2, 0.4\}$ of edges is randomly removed from the observed temporal network before inference, simulating incomplete contact tracing records. In the second setting, a fraction $\rho_n \in \{0.1, 0.2, 0.4\}$ of node states is masked, simulating scenarios where not all individuals have been tested and their SIR status is unknown. For both perturbation types, all metrics from Section 6.2 will be reported as a function of ρ_e and ρ_n respectively.

We expect the BN's edge texture representation to be particularly sensitive to edge removal, since the texture of an edge—which encodes the full sequence of contact times—is entirely lost when the edge itself is missing; this disrupts the kernel aggregation that the BN relies upon to identify backtracking paths. The DBGNN, by contrast, constructs its De Bruijn representation from the set of observed causal walks, and the dominant causal paths may survive even moderate edge removal if the network contains redundant routes. The Resistance Score $\mathcal{R}$ will additionally be computed for all architectures across the perturbed parameter grid $\mathcal{P}$, and we anticipate that architectures encoding structural rather than parametric features of the spreading process will yield the highest values of $\mathcal{R}$, reflecting a greater degree of separation between the inductive bias of the model and the specific values of β and μ assumed during training.

7

Discussion

7.1 Interpretation of Results

A recurring observation across well-performing source detection models is that the learned node representations implicitly capture the centrality of a candidate within the infected subgraph. This is consistent with the findings of [**?**], who show analytically and empirically that the true source tends to occupy a position near the topological centre of the observed infected component—a property that message-passing aggregation naturally approximates through iterated neighbourhood summation. When node features encode local infection density (the fraction of infected neighbours at observation time T), the GNN effectively learns a smoothed version of rumour centrality, weighting candidate nodes by how well their immediate neighbourhood matches the expected spreading pattern. The static GNN already exploits this signal; the question addressed by the temporal architectures is whether finer-grained information can be extracted from the contact sequence itself.

The Backtracking Network (BN) of [**?**] introduces an explicit susceptibility constraint by encoding, for each directed edge (u, v), only those timesteps at which u was infected and v was susceptible—the so-called edge texture. This expert-knowledge mask is a powerful inductive bias: it reduces the feature space to causally plausible transmission events and suppresses noise from contacts that could not have contributed to the observed outbreak. Our experiments confirm that this constraint accounts for a substantial share of the BN's advantage over the static baseline; without it, the temporal signal is diluted by the large number of non-causal contacts, particularly in dense networks with high contact rates.

Temporal structure contributes a second, complementary mechanism: causal path pruning. On a temporal contact network, the set of nodes that could realistically have been infected by a given source is strictly smaller than the set reachable on the aggregated static graph, because only time-respecting paths are epidemiologically valid. The De Bruijn GNN (DBGNN) [**?**] encodes this causal reachability at order k by lifting the network to a higher-order De Bruijn graph whose nodes represent length-k walks, thereby making non-Markovian path dependencies explicit in the graph topology. Models that can access this representation are, in principle, better positioned to eliminate candidates that lie on temporally inconsistent paths—even if those candidates appear central in the static projection.

7.2 Advantages of Temporal Architectures over Static Models

Hypothesis H1 (Causal Path Integrity) predicts that the marginal gain from temporal architectures is greatest in networks where temporal ordering strongly restricts the reachable set of nodes. This condition is most readily satisfied in sparse, structured networks—such as regular lattices or low-density Erdős–Rényi graphs—where many candidate sources can be ruled out on the basis of causal path analysis alone. In dense networks, the aggregated and temporal reachability sets converge, and the structural advantage of DBGNN or the DAG-GNN diminishes. Hypothesis H2 (Granularity Convergence) captures the complementary observation that as the time resolution of the contact data decreases—either through coarser binning or through a reduction in the observation window—the temporal architectures lose access to the fine-grained sequential information that distinguishes them from the static baseline, and their accuracy converges toward that of the static GNN. Together, H1 and H2 define a regime in which temporal methods are most valuable: sparse or structured networks observed at high temporal resolution over a sufficiently short time window that the infected component has not yet saturated the network.

Hypothesis H3 (Temporal Signal-to-Noise Resilience) addresses a distinct failure mode that affects all source detection methods at late observation times: as the fraction of infected nodes grows, the combinatorial explosion of plausible source candidates makes accurate identification increasingly difficult. We argue that temporal architectures are better equipped to handle this regime because they can leverage the ordering of infection events to partially reconstruct the transmission tree, whereas static models can only observe the final infection pattern without any information about the sequence in which it was produced. The DAG-GNN on Temporal Event Graphs is particularly well suited to this task, as the TEG representation [**?**] encodes inter-event dependencies directly in the graph topology, allowing the model to propagate information backwards along plausible transmission chains.

The three temporal architectures evaluated here occupy distinct positions on the expressiveness–cost trade-off. The BN achieves its temporal sensitivity through lightweight edge feature engineering with minimal overhead relative to the static GNN, making it the most practical choice when computational resources are limited. The DBGNN is strictly more expressive in its representation of higher-order path dependencies, but the De Bruijn graph grows combinatorially with the order parameter k, imposing substantial memory and runtime costs on large or dense temporal networks. The DAG-GNN on the TEG lies between these extremes in practice: the event graph construction is linear in the number of temporal contacts, but the resulting DAG can be large when contact rates are high, and the directed acyclic convolution has complexity proportional to the longest path in the DAG. Selecting the right architecture therefore requires balancing the structural properties of the target network against the available computational budget.

7.3 Ablation Studies

Several targeted ablations are necessary to isolate the contribution of each architectural component and to validate the three research hypotheses empirically. For the BN, the most informative ablation is the removal of the edge texture mask: replacing the susceptibility-constrained activation pattern with a naive temporal edge feature (e.g., a contact indicator vector) allows us to quantify how much of the BN's advantage is attributable to the epidemiological prior rather than to temporal information per se. Analogously, for the DBGNN, varying the De Bruijn order $k \in \{1, 2, 3\}$ controls the depth of path dependency that the model can represent; comparing performance across orders reveals whether the source detection task requires genuinely higher-order structure or whether first-order (Markovian) temporal information is sufficient. Both of these ablations bear directly on H1.

To test H2, one must vary the temporal resolution of the input by merging consecutive time bins and observing the resulting degradation in accuracy for each architecture. A model that loses its advantage over the static baseline as bin width increases provides evidence that its gains derive from fine-grained temporal ordering rather than from some coarser feature of the contact sequence. To test H3, the observation time T should be varied within each experimental setting, and performance should be tracked as a function of the fraction of infected nodes at observation time. Complementary ablations on network density —achieved by varying the edge probability in synthetic Erdős–Rényi graphs or the rewiring probability in Watts-Strogatz networks—further distinguish the regimes identified in H1 and H2 and help characterise the conditions under which each architecture provides the greatest practical benefit.

8

Limitations

The results reported in this thesis are obtained exclusively from synthetic epidemic simulations governed by the SIR compartmental model. While the SIR framework is a well-established abstraction for a broad class of infectious diseases, it makes several simplifying assumptions that may not hold in practice: infection probability is constant along each contact edge and does not vary with dose or duration, every individual recovers with equal probability per unit time, and there is no latency period between exposure and infectiousness. Consequently, GNN models that learn to exploit the statistical regularities of SIR dynamics—such as the characteristic shape of the infected component at a given recovery ratio—may not generalise to outbreaks driven by SEIR, SEIRS, or otherwise more complex dynamics. Performance on real contact-tracing datasets, where the true source, spreading rate β, and recovery rate μ are all unknown, could differ substantially from the benchmarks reported here. Validation against real-world outbreaks with a known index case, such as carefully documented laboratory or household transmission studies, is therefore an important step that lies outside the scope of the present work.

Scalability is a second significant limitation, affecting the two more expressive temporal architectures in particular. The De Bruijn graph at order k has a node set whose size grows as $O(|E|^k)$ with the number of temporal contacts $|E|$, and the De Bruijn GNN (DBGNN) [**?**] was evaluated only on networks with at most a few thousand nodes and tens of thousands of temporal contacts. The Temporal Event Graph (TEG) underlying the DAG-GNN can contain $O(|E|^2)$ directed edges in the worst case, since every pair of temporally adjacent contacts involving a shared node generates a link. Neither architecture has been stress-tested on the large-scale contact networks that arise in hospital surveillance or mobile-phone proximity studies, where $|E|$ may reach millions. The static GNN and BN are considerably more computationally tractable, but even these models face memory bottlenecks when the number of Monte Carlo simulation runs used to construct node features is large.

All four architectures share the single-source assumption: they assign a probability to each node under the hypothesis that exactly one patient-zero initiated the outbreak. This assumption simplifies training and evaluation but is routinely violated in practice—many real-world epidemics are seeded by multiple independent introduction events, particularly at the early stages of an emerging pandemic. Models trained under the single-source assumption have no mechanism to represent or detect such scenarios and will produce

misleading rankings when the true source set has cardinality greater than one. Finally, while we follow the evaluation protocol recommended by [**?**] to mitigate inflated comparisons between learned methods and the Soft-Margin Estimator baseline—which can achieve near-oracle performance when the simulation model matches the data-generating process exactly—the comparison remains constrained by the finite simulation budget, and differences between architectures that are small in absolute terms should be interpreted with care.

9

Future Work

The most pressing extension of the present work is the generalisation to multi-source detection. All architectures evaluated in this thesis assume that a single patient-zero initiated the outbreak, an assumption that is often violated during the early stages of an emerging epidemic when multiple individuals may independently introduce the pathogen into the contact network. A principled multi-source formulation requires either a multi-label output head that assigns independent probabilities to each node in a candidate source set $\mathcal{S}^*$, or a sequential detection scheme in which sources are identified and removed iteratively, analogous to peeling algorithms in combinatorial inference. Either approach demands a revised training procedure and evaluation protocol, since standard top-k accuracy metrics conflate accuracy with set-size sensitivity when $|\mathcal{S}^*| > 1$. Alongside the multi-source extension, the inductive generalisation capacity of all evaluated models deserves attention: currently, models are trained and evaluated on the same network topology, precluding deployment on previously unseen contact networks without retraining. Integrating per-node memory modules in the style of Temporal Graph Networks (TGN) [**?**]—which maintain a continuously updated state vector for each node and update it via a gated recurrent unit as new contacts arrive—would enable a model to adapt to arbitrary network topologies at inference time, a property that is essential for real-time outbreak response applications where the contact network may change between training and deployment.

On the architectural side, the De Bruijn GNN can be extended by incorporating the Hypergeometric Pattern Analysis (HYPA) framework of [**?**]. HYPA uses a hypergeometric ensemble of randomised temporal networks as a null model and retains only those higher-order temporal patterns—represented as De Bruijn graph nodes—whose observed frequency is statistically anomalous relative to the null model. Using HYPA-filtered De Bruijn graphs as input to the GNN introduces a principled, data-driven inductive bias that suppresses spurious higher-order patterns arising from random fluctuations in the contact sequence, potentially improving both accuracy and generalisation across network families. Beyond epidemic applications, the general framework developed in this thesis—temporal network representation, Monte Carlo simulation-based training, and causal GNN inference—transfers naturally to other spreading processes on temporal graphs, including rumour and misinformation propagation in social networks, malware transmission in computer infrastructure, and cascading failures in power or transport systems. Adapting the simulation engine and loss functions to these domains would

broaden the empirical scope of the methodology considerably. A further promising direction is the integration of active querying strategies, in which the model adaptively selects a small number of additional nodes whose infection status is queried to resolve ambiguity among the top-ranked source candidates [**?**]. Such a strategy converts the purely passive inference problem studied here into a sequential decision problem and offers a principled way to improve accuracy under tight observation budgets—a practically relevant setting for epidemiological investigation, where testing resources are limited.

10

Conclusions

Identifying the origin of an epidemic outbreak from a snapshot of the infection state is a problem of both scientific and practical importance. On temporal contact networks, the task is complicated by the fact that transmission follows time-respecting causal paths—a structure that static graph representations erase. This thesis has examined whether graph neural network architectures that explicitly encode temporal information can exploit this structure to improve source detection accuracy, and under what conditions such improvements are most pronounced.

10.1 Summary of Contributions

The first contribution of this thesis is a faithful re-implementation of the Backtracking Network (BN) of [**?**] within a unified training and evaluation framework based on Monte Carlo SIR simulation. The original BN was demonstrated on a specific class of contact networks; our re-implementation generalises the edge texture encoding to arbitrary temporal networks and integrates it with the wandb-tracked experimental pipeline described in Chapter 7, facilitating direct comparison against other methods under identical data conditions.

The second contribution is the adaptation of the De Bruijn Graph Neural Network (DBGNN) [**?**] and the DAG-GNN on Temporal Event Graphs for the source detection task. Both architectures were originally proposed for classification or prediction problems on temporal networks and required non-trivial modification—in particular, the replacement of their output heads and loss functions with a node-ranking formulation appropriate for source detection, and the development of dedicated data preprocessing pipelines (De Bruijn graph construction and temporal event graph construction, respectively) that interface with the shared `TSIRData` dataclass.

The third and central contribution is the first systematic multi-architecture comparison of temporal GNN methods for epidemic source detection. By evaluating the static GNN, BN, DBGNN, and DAG-GNN under identical simulation budgets, network families, and evaluation metrics—top-k accuracy, average error distance, and reciprocal rank—we provide a principled empirical basis for architectural choice in this domain. The comparison also includes classical heuristics (Jordan center, degree centrality, uniform random, and the Soft-Margin Estimator baseline of [**?**]), enabling assessment of whether learned methods offer genuine advantages over analytical alternatives.

The fourth contribution is a robustness analysis structured around three falsifiable hypotheses. H1 (Causal Path Integrity) predicts that temporal architectures gain most in sparse, structured networks; H2 (Granularity Convergence) predicts that their advantage diminishes as temporal resolution decreases; and H3 (Temporal Signal-to-Noise Resilience) predicts that they maintain accuracy at late observation times better than static models do. The experimental results provide partial or full support for each hypothesis, qualifying the conditions under which temporal GNNs constitute the preferred approach and identifying the regimes in which the additional computational cost of higher-order representations is justified.

10.2 Limitations

The primary limitation of this work is that all results are obtained from synthetic SIR simulations. Although the SIR model captures the essential compartmental dynamics of many real infectious diseases, it abstracts away heterogeneous infectivity, latency periods, reinfection, and the imperfect contact-rate assumptions inherent in any recorded contact dataset. The models trained and evaluated here may therefore exhibit different performance characteristics when applied to real-world outbreaks where the true spreading parameters are unknown and the observed infection states are subject to reporting delays or under-ascertainment. Evaluating on real outbreaks with known ground truth—such as well-documented laboratory or household transmission studies—remains an important direction for future validation.

A further limitation concerns the scalability of the more expressive temporal architectures. The De Bruijn graph at order k has a node set of size $O(|E|^k)$ where $|E|$ is the number of temporal edges, and the Temporal Event Graph can contain $O(|E|^2)$ directed links in the worst case; neither architecture was tested on networks with more than a few thousand nodes or tens of thousands of temporal contacts. The single-source assumption shared by all evaluated models is also a practical constraint: real outbreaks frequently involve multiple simultaneous or near-simultaneous introduction events, a scenario that none of the architectures is designed to handle. Finally, while we follow the evaluation protocol of [**?**] to guard against inflated comparisons with the Soft-Margin Estimator baseline, the comparison remains limited by the simulation budget, and conclusions about the relative merit of learned versus classical methods should be interpreted with appropriate caution.

10.3 Future Work

Several natural extensions follow from the limitations identified above. The most immediate is the generalisation to multi-source detection, where the model must identify a small set $\mathcal{S}^* \subset \mathcal{V}$ of patient-zero nodes rather than a single one. This requires either a multi-label output formulation or a sequential prediction scheme in which identified sources are removed and the procedure is iterated—an approach analogous to peeling al-

gorithms in combinatorial optimisation. A related extension is inductive generalisation: all models evaluated here are transductive in the sense that they are trained and tested on the same network. Incorporating per-node memory mechanisms along the lines of Temporal Graph Networks (TGN) [**?**] would allow a model trained on one contact network to be applied to an unseen network without retraining, a capability that is essential for deployment in real-time outbreak response settings.

On the architectural side, the DBGNN can be strengthened by integrating Hypergeometric Graph Ensembles (HYPA) [**?**], which use statistical hypothesis testing to identify higher-order temporal patterns that are anomalous relative to a randomised null model. Retaining only statistically significant De Bruijn edges as an inductive bias for the GNN would reduce noise from spurious higher-order patterns and potentially improve generalisation across network families. Beyond epidemic spreading, the framework developed here is directly applicable to other information diffusion processes on temporal networks, including rumour propagation in social media, malware transmission in computer networks, and cascading failures in infrastructure systems. Adapting the simulation engine and evaluation metrics to these domains would substantially broaden the impact of the methodology. Finally, integrating active querying strategies—whereby the model requests the infection status of additional nodes to resolve ambiguity among top candidates [**?**]—offers a principled way to improve accuracy at reduced observation cost, and represents a promising bridge between the passive inference paradigm studied here and practical epidemiological investigation.

A

Appendix

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List of Figures

List of Tables